

ATMT Assignment 2

Topic: thinking about experiment design and interpreting results

Deadline: Tuesday, 24. October 2025, 14:00

Important submission info:

- **Submission format: pdf**
- Please name your submission files like this: olatusername_atmt_assignmentxx.pdf, e.g. mmuster_atmt_assignment02.pdf
- For this assignment, you are asked to submit your answers in a PDF file. Please do not hand in any other files.
- You need to submit the assignment on OLAT using the tab for assignment 2. Please hand it in on time. The submission deadline is given above. After this time, the tab will be closed.
- You are encouraged to work in groups of two. Please make sure every file clearly states **both** names. For pairs, only one person should submit the assignment.

Assignment Description

In this assignment, we focus on experiment design and interpreting results for machine translation research, specifically examining how Neural Machine Translation (NMT) models handle “false friends” when using romanized versus original script data and, in part 2, examining the effects of different tokenization schemes.

1 Romanization and False Friends: Experiment Design

Below, we describe a research experiment that could serve as a student project for a seminar on machine translation. Read the sections on “Experiment Design” and “Experiment Results”. Then, answer the questions at the end of this section.

In the lecture, you learned that the way you construct your vocabulary is very important to achieve good translation quality. You are particularly interested in how a shared vocabulary improves translation when shared between languages that have many cognates. Cognates are words that have a common etymological origin and share a similar meaning, spelling, and pronunciation, e.g., “fish” in English and “Fisch” in German. On the subword level, many such cognates (or their subwords) will be shared between the source and the target language.

You decide to investigate this phenomenon for Russian and English. Russian is typically written in the Cyrillic script, while English uses the Latin script. By romanizing Russian, i.e., mapping Cyrillic characters to the Latin script, you achieve greater subword sharing between the two languages. However, this process introduces many “false friends”, which are words that share the same or a very similar surface form but do not share the same meaning. For example, the romanized Russian **бриллиант** appears similar to the English word “brilliant” but actually translates to “diamond”.

Your hypothesis is that it is harder for the romanized NMT model to learn internally which subwords are “false friends” compared to a model trained on original script data. To evaluate this, you analyze the attention weights of both NMT models. Specifically, given Russian as a source language and English as a target language, you:

- Identify all target subwords that share the same spelling as a token in the romanized source.
- Extract the corresponding source tokens in the Cyrillic script data.
- Label each pair as either a “true friend” (correct translation) or a “false friend” (incorrect translation).
- For each target subword, extract the model’s attention weights over every subword in the corresponding source sentence.
- Determine if the highest attention weight aligns with the correct source token (indicating a “true friend”) or another token (indicating a “false friend”).

Your evaluation test set is constructed from a tweet corpus containing 564 “false friends” and 6,732 shared subwords that are translations of each other, referred to as “true friends”. Using this dataset, you apply your attention-weight-based method to predict whether each pair of subwords is a “true friend” or a “false friend” for both the romanized NMT model and the original script NMT model. You then calculate the classification accuracy for each model.

1.1 Model Details

For your experiment, you train two Russian→English NMT models using 2 million unique parallel sentences from the MultiUN corpus (a collection of translated documents from the United Nations). The first model uses the original Cyrillic script for Russian, while the second model uses romanized Russian data. Both models are trained with identical hyperparameters, following standard NMT training practices using a state-of-the-art framework. Early stopping is employed when the BLEU score on a development set of 10,000 parallel sentences does not improve for 5 validation runs. During training, a beam size of 4 is used to speed up the process, and a beam size of 12 is used during translation. The BLEU scores are computed using sacreBLEU on the raw text.

1.2 Experiment Results

The results of your experiment are summarized in the following table:

	Cyrillic Russian	Romanized Russian
Accuracy	82.43%	60.43%

The table shows the classification accuracy of your attention-weight-based identification method for the two models: one trained on original script data (Cyrillic Russian) and one trained on romanized data (Romanized Russian).

1.3 Submission

Submission: Answer the following questions in your PDF report with a short paragraph each. **Don’t forget that there’s another experiment and set of questions to answer after this one.**

- Discuss the results for the experiment in detail. What does the table show?
- Did you expect this outcome?
- What conclusions can you draw from the results?
- Are there any other experiments you could run to further back up your conclusions?

2 Tokenization and Interpreting Perplexity Scores: Experiment Design

MorphPiece is a tokenization method that combines linguistic knowledge with statistical approaches to create a subword tokenization scheme. At its core, MorphPiece is based on morphological segmentation, which breaks words down into their constituent morphemes - the smallest units of meaning in a language. This linguistic foundation is derived from MorphyNet, a database of derivational and inflectional morphology for multiple languages.

MorphPiece follows a hybrid approach to tokenization in which it combines two strategies: morphological segmentation using a lookup table (MorphTable) constructed from MorphyNet, and Byte Pair Encoding (BPE) for words not found in the MorphTable. This dual approach allows MorphPiece to leverage linguistic knowledge where available, while still maintaining the flexibility to handle novel or rare words.

The tokenization process begins with standard text normalization and pre-tokenization, similar to BPE. For each pre-token, MorphPiece first checks if it exists in the MorphTable. If found, the word is segmented into its constituent morphemes. If not, it's tokenized using a custom-trained BPE vocabulary.

Word	BPE tokens	MorphPiece tokens
batting	'bat', 'ting'	'bat', '#ing'
disengage	'dis', 'eng', 'age'	'dis#', 'en#', 'gage'
archeologists	'ar', 'che', 'ologists'	'archaeo#', '#logy', '#ist', '#s'
eyewitnesses	'ey', 'ewitness', 'es'	'eye', '#', 'wit', '#ness', '#s'
photographers	'phot', 'ographers'	'photo#', 'graph', '#er', '#s'

The final vocabulary in MorphPiece consists of unique affixes and stems from the MorphTable, supplemented by additional tokens from BPE training. The total vocabulary size is kept comparable to standard models (around 50,000 tokens). Special tokens are used to mark prefixes, suffixes, and compound words, providing additional structural information to the model.

One challenge with MorphPiece is that it requires a more complex detokenization process due to its hybrid nature. It uses a reverse MorphTable and heuristics to reconstruct words from morphemes, which is more involved than the simple concatenation used in BPE detokenization.

While this hybrid approach often results in slightly longer sequences, its authors hope it may lead to better semantic representations, especially for morphologically rich languages. In experiments, models using MorphPiece tokenization exhibit lower perplexity scores across various datasets.

Tokenizer	Average Length	Fertility
Whitespace	526.69	N/A
BPE	585.30	1.111
MorphPiece	685.78	1.302

Note: Average Length and Fertility comparison of different tokenizers. Whitespace tokenization is a simple method that splits text into tokens at whitespace characters (spaces, tabs, newlines). It's often used as a baseline in natural language processing. Fertility, in tokenization, refers to the average number of tokens produced per word.

Metric	GPT-2 Base	Morph 200k	GPT-2 Large
PennTreeBank	61.859	38.251	37.931
OpenAI-250k	25.544	17.262	16.743

Note: The metric is perplexity (lower is better). Remember that, while Morph 200k uses the MorphPiece tokenizer, GPT-2 uses traditional BPE tokenization.

To understand the results, it's necessary to understand the concept of perplexity, a common metric used in evaluating language models. Perplexity measures how well a probability model predicts a sample. In the context of language models, it quantifies the model's uncertainty in predicting the next word or token. Mathematically, perplexity is calculated as the exponential of the average negative log-likelihood of a sequence:

$$\text{Perplexity} = \exp(-1/N * \sum \log P(\text{token}_i | \text{context}))$$

where N is the number of tokens in the sequence. A lower perplexity score suggests the model is more certain about its predictions.

Interestingly, perplexity can be thought of as similar to a branching factor in decision trees or search algorithms. In this analogy, perplexity represents the average number of equally likely choices the model faces when predicting the next token. For instance, a perplexity of 100 suggests that the model's performance is equivalent to randomly guessing each token from 100 equally probable options.

2.1 Submission

Submission: Answer the following questions in your PDF report with a short paragraph each:

- Discuss the results for experiment 3. What does the table show about the performance of MorphGPT compared to GPT-2 Base and GPT-2 Large?
- Is the perplexity metric impacted by different tokenization techniques? How might this affect the interpretation of the results in the table?
- MorphPiece typically produces longer sequences than BPE. What is the impact of longer sequences on perplexity scores, everything else being equal? How does this relate to the results we see for MorphGPT?
- Based on these results and your understanding of tokenization methods, what further experiments would you propose to investigate the strengths and limitations of MorphPiece tokenization?

Please let us know in the OLAT forum if you have problems or something is unclear.
Good luck!